```
#include <stdio.h>
#include <pthread.h>
#include <unistd.h>
#define N 5
pthread_mutex_t forks[N];
pthread_t philosophers[N];
void* philosopher(void* num)
{
    int id = *(int*)num;
    int left = id;
    int right = (id + 1) % N;
    while (1)
    {
        printf("Philosopher %d is thinking.\n", id);
        sleep(1);
        if (id % 2 == 0)
        {
            pthread_mutex_lock(&forks[left]);
            printf("Philosopher %d picked left fork %d\n", id, left);
            pthread_mutex_lock(&forks[right]);
            printf("Philosopher %d picked right fork %d\n", id, right);
        }
        else
        {
            pthread_mutex_lock(&forks[right]);
            printf("Philosopher %d picked right fork %d\n", id, right);
            pthread_mutex_lock(&forks[left]);
            printf("Philosopher %d picked left fork %d\n", id, left);
        }
        printf("Philosopher %d is eating.\n", id);
```

```
        sleep(2);
        pthread_mutex_unlock(&forks[left]);
        pthread_mutex_unlock(&forks[right]);
        printf("Philosopher %d put down forks %d and %d\n", id, left, right);
    }
    return NULL;
}
int main()
{
    int i;
    int ids[N];
    for (i = 0; i < N; i++)
    {
        pthread_mutex_init(&forks[i], NULL);
        ids[i] = i;
    }
    for (i = 0; i < N; i++)
    {
        pthread_create(&philosophers[i], NULL, philosopher, &ids[i]);
    }
    for (i = 0; i < N; i++)
    {
        pthread_join(philosophers[i], NULL);
    }
    for (i = 0; i < N; i++)
    {
        pthread_mutex_destroy(&forks[i]);
    }
    return 0;
}
```

```
Philosopher 1 is thinking.
Philosopher 0 is thinking.
Philosopher 4 is thinking.
Philosopher 2 is thinking.
Philosopher 3 is thinking.
Philosopher 3 picked right fork 4
Philosopher 3 picked left fork 3
Philosopher 3 is eating.
Philosopher 2 picked left fork 2
Philosopher 0 picked left fork 0
Philosopher 0 picked right fork 1
Philosopher 0 is eating.
Philosopher 0 put down forks 0 and 1
Philosopher 0 is thinking.
Philosopher 2 picked right fork 3
Philosopher 2 is eating.
Philosopher 3 put down forks 3 and 4
Philosopher 3 is thinking.
Philosopher 4 picked left fork 4
Philosopher 4 picked right fork 0
Philosopher 4 is eating.
Philosopher 2 put down forks 2 and 3
Philosopher 4 put down forks 4 and 0
Philosopher 4 is thinking.
Philosopher 0 picked left fork 0
Philosopher 1 picked right fork 2
Philosopher 2 is thinking.
Philosopher 3 picked right fork 4
Philosopher 0 picked right fork 1
Philosopher 0 is eating.
Philosopher 3 picked left fork 3
Philosopher 3 is eating.
Philosopher 0 put down forks 0 and 1
Philosopher 0 is thinking.
Philosopher 1 picked left fork 1
Philosopher 1 is eating.
Philosopher 3 put down forks 3 and 4
Philosopher 3 is thinking.
```